Project Title:

Scrolling Less, Studying More: A Mobile-Based Productivity Tracking System for Improving Student Focus and Academic Efficiency

Project Objective or Aim:

The purpose of this research is to examine how a mobile-based productivity tracking system can improve focus and academic efficiency among college students. The study will investigate whether real-time tracking of screen time, app usage, and study duration can positively influence student behavior and reduce distractions. The main research question is: How does real-time productivity tracking impact study habits and focus levels in college students? This project aims to design and evaluate a prototype mobile application that provides students with feedback on their study sessions and digital distractions. The findings will help determine whether such tools can improve time management and academic performance in a university environment.

Project Background and Significance:

College students face increasing challenges in maintaining focus due to constant digital distractions, particularly from smartphones and social media platforms. Research has shown that excessive screen time and frequent interruptions can negatively affect cognitive performance, attention span, and academic outcomes. While productivity tools exist, many fail to provide real-time feedback or personalized insights that actively influence behavior during study sessions.

This project is significant because it addresses a growing issue within higher education: the inability of students to effectively manage their time in a highly digital environment. By focusing on real-time productivity tracking, this research seeks to bridge the gap between awareness and behavioral change. Instead of simply showing daily screen time summaries, the proposed system will provide immediate feedback during study sessions, allowing users to recognize and adjust their behavior as distractions occur.

The theoretical framework for this research is based on behavioral self-regulation and cognitive load theory. Behavioral self-regulation suggests that individuals can modify their actions when provided with timely and relevant feedback. Cognitive load theory explains how excessive distractions can overwhelm working memory, reducing learning efficiency. By combining these frameworks, this project aims to demonstrate how real-time data can support better decision-making and improved focus.

Additionally, this research contributes to the field of information technology by exploring how mobile applications can be designed not just for convenience, but for behavioral impact. The results may influence future development of educational technology tools that prioritize user engagement and productivity. Within the University of Central Florida (UCF) community, this project has the potential to provide students with practical solutions for managing distractions and improving academic performance, particularly during high-stress periods such as midterms and finals.

Research Methods:

This project will follow a structured approach that includes design, development, testing, and evaluation of a mobile-based productivity tracking system. The first step will involve conducting background research on existing productivity applications and identifying their strengths and limitations. This will help define the key features needed for the proposed system.

Next, a prototype mobile application will be developed using basic front-end and back-end technologies. The application will track metrics such as study session duration, app usage during study periods, and frequency of distractions. A simple user interface will be created to display real-time feedback, including alerts when users switch to distracting applications.

After development, the project will move into the testing phase. A small group of 5–10 undergraduate students will be recruited to use the application over a two-week period. Participants will be asked to complete short surveys before and after the testing phase to measure changes in focus, study habits, and perceived productivity. This step is necessary to evaluate the effectiveness of the system in a real-world setting.

The final phase will involve analyzing the collected data. Quantitative data from app usage logs will be compared with survey responses to identify patterns and correlations. For example, reduced app switching during study sessions may indicate improved focus.

Timeline (May–August):

May: Research and planning

June: Application development

July: Testing with participants

August: Data analysis and final report

Each step is essential to ensure that the system is both functional and capable of producing meaningful results.

Expected Outcome:

The expected outcome of this project is the development of a functional mobile application prototype that helps students monitor and improve their productivity during study sessions. The primary deliverables will include the application itself, a written research report, and a potential poster presentation for an undergraduate research conference at UCF.

The project is expected to demonstrate that real-time feedback can positively influence student behavior by reducing digital distractions and encouraging more focused study sessions. If successful, the results will show measurable improvements in participants' ability to stay on task and manage their time effectively.

In terms of knowledge contribution, this research will provide insights into how technology can be used to support behavioral change rather than simply track activity. It will expand on existing studies by focusing specifically on real-time intervention, rather than passive data collection. This distinction is important because it highlights the role of immediate feedback in shaping user behavior.

For the UCF community, the project offers practical value by addressing a common challenge faced by students. The findings could inform future development of campus-supported tools or applications aimed at improving academic performance. Additionally, the project may serve as a foundation for further research in areas such as user experience design, educational technology, and digital wellness.

Ultimately, this research aims to show that small technological interventions can have a meaningful impact on student success. By combining data tracking with user-centered design, the project seeks to create a tool that is both effective and accessible for everyday use.

Literature Review:

Rosen, L. D., et al. (2016). The distracted student.

Junco, R. (2017). Student class standing, Facebook use, and academic performance.

Carrier, L. M., et al. (2018). Multitasking across generations.

Mark, G., Gudith, D., & Klocke, U. (2018). The cost of interrupted work.

Duke, É., & Montag, C. (2017). Smartphone addiction and productivity.

Bowman, L. L., et al. (2019). The impact of media multitasking.

Preliminary Work and Experience:

As an Information Technology major at the University of Central Florida, I have developed a strong foundation in programming, database management, and software development. My coursework has included experience with languages such as Python, C, SQL, HTML, and CSS, as well as tools like GitHub for version control and collaboration. These skills are directly applicable to the development of a mobile-based productivity tracking system.

In addition to technical skills, I have participated in group projects that involved designing and troubleshooting applications, which has strengthened my ability to work through real-world technical challenges. I have also gained experience in organizing project components, managing workflows, and refining ideas through collaboration.

Although this project represents a new area of research for me, my background in technology and problem-solving provides a strong foundation for success. I am confident in my ability to learn new tools and apply my knowledge to develop a functional and effective prototype.

IRB/IACUC Statement:

This project involves human participants through surveys and application testing; therefore, IRB approval will be required prior to data collection.

Budget:

Participant incentives ($10 × 10 students): $100

Software/tools (development resources): $200

Cloud hosting/database services: $300

Equipment/testing devices: $500

Miscellaneous expenses: $400

Total: $1,500